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Review Article

Unmanned aerial vehicles for communication recovery in post-disaster scenarios: A PRISMA-based systematic review

Raymond Thapa Magar [a], Tien Phat Dinh [b], The Hung Nguyen [b], Quang Hoai Le [c,*], Ahn Yonghan [d]

[a] Department of Applied Artificial Intelligence, Hanyang University-ERICA, Ansan, 15588, Republic of Korea
[b] Department of Smart City Engineering, Hanyang University-ERICA, Ansan, 15588, Republic of Korea
[c] Center for AI Technology in Construction, Hanyang University-ERICA, Ansan, 15588, Republic of Korea
[d] Department of Architectural Engineering, Hanyang University-ERICA, Ansan, 15588, Republic of Korea

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ABSTRACT

Rapid restoration of communication infrastructure is essential for effective coordination, emergency response, and resource allocation in post-disaster scenarios. As conventional communication systems are often damaged or rendered inaccessible during disasters, Unmanned Aerial Vehicles (UAVs) have emerged as a promising alternative for establishing temporary, ad-hoc communication networks. However, despite growing academic and practical interest, UAV-based communication recovery remains in its early stages, facing numerous technical and operational challenges that hinder widespread and effective adoption in complex disaster environments. In response, this study conducts a systematic literature review following the PRISMA protocol to synthesize the current state of knowledge on UAVs applications for Aerial Base Stations (ABSs) deployed as temporary replacements for disaster-affected terrestrial base stations. The review identified key concepts and system architectures, hardware configurations, deployment strategies, emerging AI-driven autonomy, and energy optimization approaches. The findings reveal notable technological progress in energy harvesting (e.g., solar-assisted and tethered platforms) and AI-driven energy conservation strategies; however, persistent limitations, particularly sustained endurance and unresolved coverage–capacity trade-offs, continue to hinder real-world implementation due to the lack of robust real-world validation. By consolidating these insights, the study provides a comprehensive assessment of current advancements and identifies strategic priorities for future research and policy development. This plays a crucial role in supporting the design and deployment of resilient, adaptive, and efficient UAV-based ABSs systems in disaster management contexts.

1. Introduction

In recent decades, natural disasters, exacerbated by climate change, have increasingly resulted in substantial loss of life, large-scale population displacement, and widespread socio-economic disruption, thereby posing severe threats to human safety and societal stability [1–4]. In addition to the immediate human toll, these events severely damage critical urban infrastructure, including housing, transportation systems, and especially communication networks. The failure of such systems hampers emergency response and delays recovery efforts [5,6]. Disruptions to essential services, such as transportation, telecommunications, power supply, and water, can limit access to mobile networks, electricity, and the internet, further undermining disaster response capabilities [7].

Communication systems are fundamental to modern societies, and their importance becomes paramount in post-disaster scenarios. Rapid restoration of communication infrastructure is essential for coordinating emergency responses, disseminating life-saving information, and maintaining links between affected communities and relief agencies [8,9]. Reliable networks enable first responders, authorities, and humanitarian organizations to assess damage, allocate resources efficiently, and deliver aid effectively [9,10]. However, restoring conventional communication systems after disasters is often hindered by significant challenges [10]. Infrastructure may be destroyed or inaccessible due to debris, difficult terrain, or ongoing hazards, while reconstruction efforts are delayed by logistical constraints, limited access to remote areas, and

* Corresponding author.
E-mail address: hoaile10@hanyang.ac.kr (Q.H. Le).

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the scale of the damage. These challenges underscore the need for alternative, resilient communication strategies that can be rapidly deployed to ensure temporary yet crucial connectivity during the early stages of disaster recovery [3,8].

As the frequency and impact of natural disasters continue to rise, there is growing recognition of the need for innovative technologies to enhance urban resilience and disaster preparedness [11,12]. Recent advancements in digital and autonomous systems have opened new opportunities to support emergency response operations, especially in scenarios where conventional infrastructure has been compromised [13]. Among these emerging technologies, Unmanned Aerial Vehicles (UAVs) have gained considerable attention for their potential to serve as flexible, rapidly deployable platforms in disaster management contexts [7,8,14]. In particular, UAVs have demonstrated promising capabilities in restoring communication systems during the critical early stages of disaster recovery [4]. Their mobility and adaptability, proving advantageous in various applications [15,16], enable them to access remote or hazardous areas where ground-based reconstruction is delayed or infeasible. Equipped with lightweight communication payloads, UAVs can act as temporary airborne relays or base stations, facilitating emergency communication and coordination among rescue teams, government agencies, and affected communities [7,8].

Despite these advantages, the integration of UAVs into post-disaster communication restoration remains in an early stage of development [17]. Although academic interest has increased, a clear research gap persists. Existing reviews are often overly broad, either surveying UAVs applications across diverse disaster management tasks (e.g., mapping, search and rescue, and logistics) or aggregating heterogeneous communication technologies, including terrestrial, aerial, and space-borne networks [18,19]. In contrast, more narrowly focused studies typically emphasize auxiliary or protocol-level use cases, rather than addressing base-station-level network replacement [20,21].

Consequently, there is a lack of systematic synthesis centered on UAV-based Aerial Base Stations (ABSs) deployed as temporary replacements for non-operational terrestrial base stations, a deployment paradigm that differs fundamentally from short-term emergency coverage or incremental capacity enhancement. This gap is critical because replacement-level deployments must contend with persistent technical and operational constraints, including limited endurance, environmental vulnerability, regulatory compliance, backhaul availability, and the absence of standardized deployment protocols [3,4,8,14]. These challenges are further amplified in low-resource or high-risk settings, where infrastructure and trained personnel are limited. Accordingly, a focused systematic review restricted to empirical studies is needed to consolidate evidence and inform practical deployment strategies for post-disaster communication recovery.

Motivated by the identified gap, this study addresses the need to consolidate technical and operational knowledge on UAV-based communication restoration in post-disaster settings. The review focuses specifically on UAV-based ABSs deployed as temporary replacements for non-operational terrestrial base stations, rather than auxiliary or short-term connectivity solutions. It synthesizes key operational dimensions of replacement-level deployments, including platform design, endurance and power management, backhaul integration, deployment strategies, and emerging AI-enabled autonomy and optimization, together with their associated constraints. Accordingly, this study conducts a systematic review to (1) synthesize validated system architectures and deployment models, (2) evaluate their feasibility, benefits, and limitations, and (3) identify critical technical and operational barriers, thereby informing future research directions, deployment frameworks, and standardization efforts.

To ensure methodological rigor and transparency, the review follows a systematic literature review methodology guided by the PRISMA (Preferred Reporting Items for Systematic Reviews and Meta-Analyses) protocol [22]. The resulting synthesis aims to consolidate fragmented evidence, map research trends, and identify underexplored areas in UAV-assisted disaster communication. In doing so, it provides actionable insights for policymakers, emergency planners, and technology developers regarding the operational viability of UAV-based ABSs, and supports the development of standardized deployment protocols and frameworks, particularly for resource-constrained and high-risk regions where rapid and resilient communication recovery is most critical.

2. Methodology

A systematic literature search was conducted using Web of Science and IEEE Xplore as the primary databases. The search was performed in December 2025 to identify empirical studies on UAV-based ABSs for post-disaster communication restoration. Search queries were formulated using Boolean operators and combined terms related to UAVs platforms, aerial base stations, and disaster-driven communication needs, including: ("Unmanned Aerial Vehicle" OR "UAV" OR "Drone" OR "Aerial Base Station" OR "ABS") AND (communication* OR coverage*) AND (emergency* OR disaster* OR earthquake OR storm).

To ensure relevance and methodological rigor, predefined inclusion and exclusion criteria were applied (Table 1). Studies were included if they: (i) were published in English; (ii) appeared as peer-reviewed journal articles; (iii) were published in the last decade, from 2015 to 2025; and (iv) presented empirical system-level evidence (simulation-based evaluation, experimental validation, field trials, or real-world deployments) in post-disaster or large-scale infrastructure failure contexts. Non-empirical works (e.g., reviews, editorials, or purely conceptual studies) were excluded.

The study selection followed a two-stage screening process. First, titles and abstracts were independently screened to exclude studies clearly outside the review scope. Second, full texts of potentially eligible articles were assessed against the eligibility criteria. The overall selection process and reasons for exclusion at each stage are documented in the PRISMA flow diagram (Fig. 1). The initial database search identified 2060 records. After removing duplicates, 1,883 records for further screening. Manual screening of titles, abstracts, and keywords subsequently excluded 1,742 records, comprising 1,708 irrelevant studies and 34 review papers. This process yielded 141 studies for full-text assessment, of which one could not be retrieved. The remaining 140 studies were evaluated for eligibility, resulting in the exclusion of 35 records that were limited to conceptual or purely theoretical frameworks. In addition, 12 relevant studies were identified through cross-referencing, leading to a final corpus of 117 eligible studies for in-depth analysis and

Table 1
Eligible criteria.

Category	Criteria
Included Studies	• Post-disaster or large-scale infrastructure failure scenarios with partially or fully inoperative terrestrial cellular networks (e.g., earthquakes, floods, landslides, hurricanes, catastrophic outages) • UAV-based ABSs deployed as temporary replacements for non-operational terrestrial base stations • Provision of cellular-equivalent services (e.g., LTE/5G radio access, user connectivity, network operation) • Empirical system-level evidence, including simulation-based evaluations, experimental validations, field trials, or real-world deployments • Peer-reviewed journal articles published between 2015 and 2025
Excluded Studies	• Routine, planned, or non-disruptive contexts (e.g., urban optimization, capacity planning, network densification) • UAVs deployments limited to coverage extension, traffic offloading, or capacity enhancement of operational terrestrial networks • Purely theoretical, conceptual, or analytical studies lacking simulation, experimental validation, or deployment-based assessment • Non-empirical works (e.g., reviews, editorials, etc.)

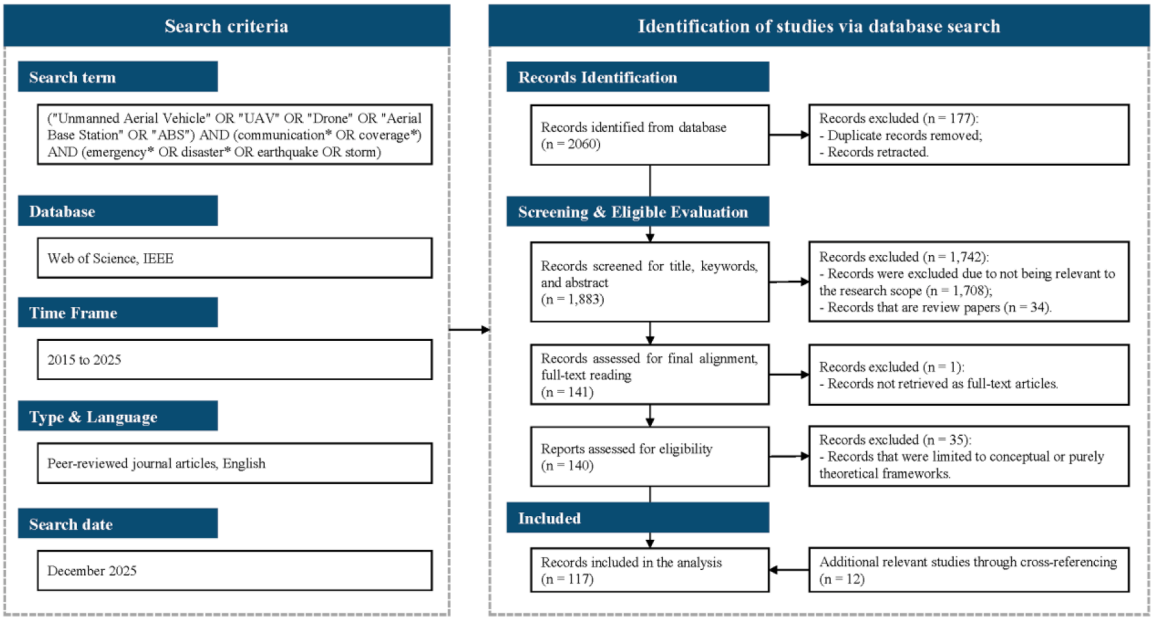


Fig. 1. PRISMA diagram of data collection for the literature review.

synthesis.

3. Results

3.1. General observations

Fig. 2 illustrates the annual trends in publications and citations related to the use of UAVs as ABSs. Inspection of the figure shows that research activity began to emerge in 2018, followed by a steady increase in publications, reaching 13 papers in 2020. During this period, academic interest also intensified, as annual citations rose sharply to 3,326. Although the number of publications declined to five in 2021, research output subsequently rebounded, reaching 26 papers in 2023. Although the annual output saw a slight decrease in the following years, falling to 25 papers in 2024 and 19 in 2025, the academic impact continued its upward trajectory. Annual citations achieved their highest point at over 4,891 by 2025.

Table 2 summarizes the geographical distribution of research on UAV-based communication recovery in post-disaster scenarios by region and country. The results reveal a strong regional concentration, with Asia dominating the literature through 84 publications, led by China with 42 papers; nearly four times the output of the United States (11 papers), followed by India and South Korea (8 papers each). Additional contributions from Japan (5) and Malaysia, Pakistan, and Turkey (4 each) further reflect Asia's strong engagement in UAVs technologies for disaster-prone contexts. Europe contributed 14 studies, primarily from the UK (4), Spain, and Italy (2 each), while North America accounted for 12 publications, largely from the USA (11). Smaller but notable contributions were observed from South America and Africa, including Brazil (3), Chile, Colombia, Ecuador, and Tunisia (1 each). Overall, while UAV-based communication recovery research is geographically diverse, it remains concentrated in a small number of countries, with China emerging as the leading contributor, underscoring regional priorities and the growing global recognition of UAVs in disaster management.

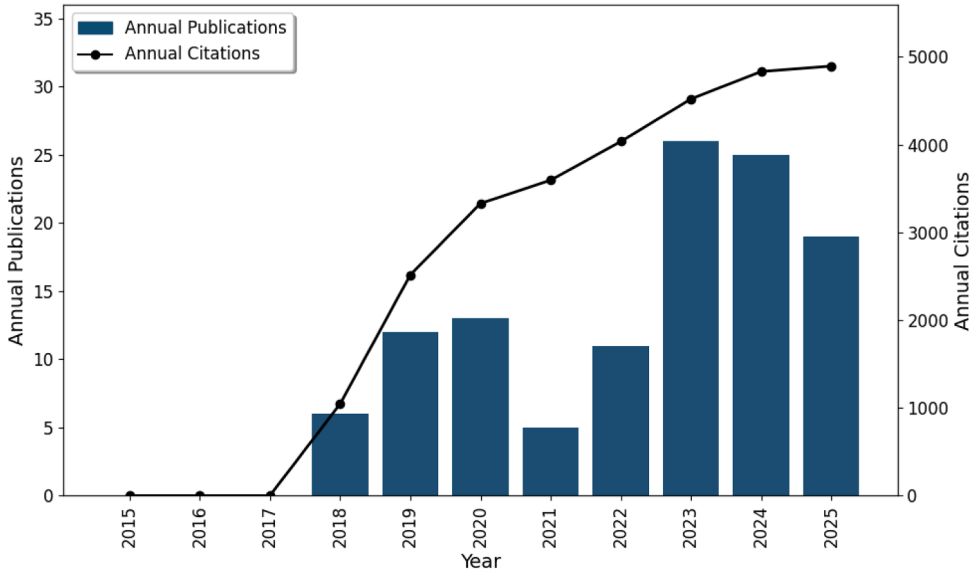


Fig. 2. Publications and citations trend by year.

Table 2
Geographical distribution of included studies.

Continent (count)	Country	Article	Continent (count)	Country	Article
Asia (84)	China	42	Europe (14)	UK	4
	India	8		Spain	2
	South Korea	8		Italy	2
	Japan	5		Czech Republic	1
	Malaysia	4		Ireland	1
	Pakistan	4		Luxembourg	1
	Turkey	4		Netherlands	1
	Iran	3		Slovakia	1
	Saudi Arabia	3		Sweden	1
	Bangladesh	1	South America (6)	Brazil	3
	Jordan	1		Chile	1
	Sri Lanka	1		Colombia	1
North America (12)	USA	11		Ecuador	1
	Canada	1	Africa (1)	Tunisia	1

3.2. UAVs-based communication recovery in post-disaster scenarios

In post-disaster scenarios, restoring communication is critical, and UAVs have emerged as a promising technological solution to support this effort. Their high mobility, rapid deployability, and growing payload capacities make them well-suited to function as ABSs, temporarily replacing damaged ground infrastructure to provide wireless coverage in affected areas [23,24]. This capability is especially vital during the response phase of the disaster management cycle, when timely communication is essential for operational coordination and saving lives [25]. UAV-based communication systems address these challenges by serving as aerial platforms capable of swiftly re-establishing connectivity [26].

These systems typically operate in two roles: as Airborne Relays (AR) and as ABS. In the AR role, UAVs function as communication bridges, relaying signals between user devices (e.g., mobile phones or emergency radios) and distant operational ground base stations [27,28]. UAVs in this role can dynamically adjust their trajectories and schedules to support multi-hop communication, particularly in remote or isolated regions. In contrast, UAVs in the ABS role are equipped with broadcasting hardware such as femtocells or picocells, enabling them to deliver direct cellular or Wi-Fi coverage to users within a defined area [29,30]. These UAV-ABS systems can rapidly establish wireless access points, facilitating real-time communication and enhancing coordination during rescue efforts [5,31]. The key advantages of UAV-based communication systems include their rapid deployment, high mobility, and ability to maintain line-of-sight (LoS) communication links [17,32]. Unlike conventional terrestrial systems that require extensive time and resources to restore, UAVs can be deployed almost immediately, providing critical communication services when they are needed most [33,34].

The effective operation of a UAVs-based communication system hinges on the seamless integration of several key components, including the UAVs platform itself, its communication payload, the navigation system, and the control architecture. First, the physical platform can be a rotary-wing UAVs, which offers vertical take-off and landing capabilities suitable for spatially constrained areas, or a fixed-wing UAVs, which provides longer flight endurance and broader area coverage, ideal for large-scale missions [16,35], or a hybrid model that combines the advantages of both for enhanced deployment flexibility [36]. Second, the communication payload comprises transceivers, antennas, and compact base stations like femtocells or picocells [27,32]. Third, the navigation and guidance system ensures the UAVs can accurately determine their position and maneuver effectively [37]. The Global Positioning System is a primary component, providing real-time location data [36], often augmented by an Inertial Measurement Unit to enhance precision and stability, especially in noisy or urban environments. Finally, the control system encompasses a Ground Control Station (GCS), data links, and control algorithms [38]. The GCS provides a human-machine interface for monitoring, trajectory planning, and commanding the UAVs [39]. The tight integration of these components enables a UAV-based communication system to function effectively in emergencies.

Beyond the physical components, the successful deployment of UAVs-based ABSs is contingent upon meeting stringent technical requirements to ensure effectiveness and reliability in harsh environments [17,25]. Foremost among these is the need for robust connectivity and coverage, which enables effective communication in disaster-stricken areas [27]. The UAVs system must provide high throughput and low latency to support real-time applications, such as live video streaming or rescue coordination [29,40]. Furthermore, it must guarantee wide and stable coverage for many users, particularly in densely populated or isolated zones. Previous research indicates that optimizing UAVs altitude and placement can maximize the coverage area while minimizing interference and signal loss [30,32]. A second critical requirement is endurance and energy efficiency, which are necessary to support long-duration missions, especially in post-disaster scenarios that may last for days [16]. A third requirement is stability and reliability, ensuring the system's operational integrity under complex environmental conditions [25,41]. Platform-wise, the UAVs must maintain stability against factors like strong winds or rugged terrain through advanced control systems and inertial sensors [37]. Maintaining a strong and uninterrupted communication link is essential, especially in environments with high interference or signal degradation [32]. To enhance reliability, UAVs networks are often designed with self-healing capabilities, allowing the network to dynamically adjust when a UAVs fails, through distributed algorithms or swarm coordination [31,42]. Security and safety are equally important considerations. UAVs are vulnerable to attacks such as jamming or spoofing, which can disrupt rescue operations [43]. It is therefore imperative to apply robust encryption protocols and anti-jamming techniques, such as using software-defined radio to adapt to threats [34]. Moreover, the integrity and confidentiality of transmitted data must be ensured via advanced encryption methods like end-to-end encryption to prevent unauthorized access in sensitive scenarios [5,38]. Together, these requirements form the basis of a robust and dependable UAVs communication system for disaster management.

An analysis of the reviewed literature reveals a varied landscape in methodological classification, with studies spanning from conceptual models to practical deployments. The body of work primarily focuses on developing technological frameworks, optimizing deployment, and integrating advanced technologies like 5G and machine learning to enhance communication effectiveness in emergency scenarios [27,33]. Table 3 reveals a significant disparity across the levels of research maturity, as studies focusing on simulation predominantly outweigh those centered on real-world deployment and field trial, highlights substantial barriers in translating research into practical implementation. This is also clearly reflected in the technical preferences: RWDs are the commonly examined hardware for investigation, owing to their versatility across conceptual development and applied testing, whereas HAPs remain limited in use, particularly in real-world scenarios. Deployment strategies also display uneven adoption, with centralized strategy used far more often than its decentralized and hybrid counterparts. Similarly, within the application of automation models, the path optimization phase receives the most research attention. Finally, regarding energy optimization, energy conservation solutions are far more extensively researched than harvesting methods.

3.2.1. UAVs types and capabilities for aerial base stations

UAV-based ABSs have emerged as a critical enabler of rapid and resilient connectivity restoration in post-disaster environments, where terrestrial infrastructure is often damaged or inaccessible. Among

Table 3
Distribution of UAV-based communication recovery studies by domain and research maturity.

Thematic domains		Article Count			References		
		SE	EV	FT	SE	EV	FT
Hardware	RWDs	96	3	4	[23,24,26,40,43–134]	[39,135,136]	[41,137–139]
	FWDs	6*	-	-	[92,95,110,140–142]	-	-
	HAPs	1*	1	-	[78]	[143]	-
	N.a	10	-	-	[144–153]	-	-
Deployment Strategies	Centralized	77	4	3	[23,24,26,40,44–46,48–57,59,61,63–70,72–74,77,79,81–86,89,92,94–96,98–104,106–108,111,112,115–119,121–124,126–128,130,132–134,140,144–148]	[39,135,136,143]	[41,137,139]
	Decentralized	22	-	1	[43,47,62,71,87,88,90,91,93,97,105,109,113,114,120,125,129,131,149,151–153]	-	[138]
	Hybrid	10	-	-	[58,60,75,76,78,80,110,141,142,150]	-	-
AI-driven Optimization	Path optimization	50	-	-	[23,24,26,40,43,45,46,48–54,58,59,63–65,67,71,74,76,77,81–83,86,90,92,94,98–102,104,105,110,116,121,124,127,128,130,131,133,134,148,151]	-	-
	Autonomous Control	40	1	-	[47,60,62,66,68–70,73,75,78–80,87,88,91,93,95–97,106–109,112–115,117,120,122,125,126,129,132,142,146,149,150,152,153]	[39]	-
	N.a	19	3	4	[44,55–57,61,72,84,85,89,103,111,118,119,123,140,141,144,145,147]	[135,136,143]	[41,137–139]
Energy Optimization	Conservation	72	1	-	[23,24,40,43,45,46,48,50,55–57,59–64,66,68–71,73–75,77–79,84,89,92–95,97,100,102–118,120–124,126–134,141,145,146,150,151]	[39]	-
	Harvesting	7*	2	2	[44,76,86,89,116,119,145]	[135,136]	[41,139]
	N.a	33	1	2	[26,47,49,51–54,58,65,67,72,80–83,85,87,88,90,91,96,98,99,101,125,140,142,144,147–149,152,153]	[143]	[137,138]

Note: - (*): some studies have adopted multiple types of UAVs
- SE = Simulation-based System Evaluation; EV = Experimental Validation; FT = Field Trial; RWDs = Rotary-Wing Drones; FWDs = Fixed-Wing Drones; HAPs = High-Altitude Platforms

available platforms, rotary-wing UAVs (e.g., quadcopters and hexacopters) form the cornerstone of localized emergency access due to their ability to hover, maneuver precisely, and deploy rapidly. Their stationary flight capability allows them to operate as quasi-static aerial base stations positioned directly above clusters of users, ensuring persistent and reliable coverage for emergency responders and affected populations [68,71,123]. Vertical take-off and landing (VTOL) aircraft further enables operation in confined, debris-filled urban settings without reliance on runways or launch systems [40,92,136]. Fine-grained three-dimensional positioning enhances LoS conditions and allows dynamic adaptation to shifting user distributions, making rotary-wing UAVs particularly effective within inner mission areas that demand precision and responsiveness [78,85,110]. These advantages, however, come with notable constraints: limited battery capacity and high energy consumption for lift restrict endurance to tens of minutes, resulting in small individual coverage areas and necessitating coordinated multi-UAV swarms for large-scale disasters [40,43,45,111,136]. Additional limitations include modest payload capacity and vulnerability to adverse weather, which can further reduce operational stability and service duration [40,46,83,129,135].

To complement these access-oriented platforms, fixed-wing and high-altitude UAVs assume strategic roles that emphasize wide-area coverage and backbone connectivity. Fixed-wing UAVs excel in endurance, speed, and range, enabling sustained patrol of disaster perimeters and efficient long-distance data relaying between ground base stations, aerial clusters, and remote command centers [92,110]. Their greater payload capacity supports more powerful communication equipment, reinforcing their effectiveness as mobile relay or backbone nodes when terrestrial links are compromised [92,141]. High-altitude UAVs further enhance network robustness by operating above most obstacles, significantly increasing LoS probability, reducing shadowing effects, and expanding geometric coverage footprints [140,142,143]. These benefits stabilize air-to-ground links and can improve aggregate network performance when altitude and spectrum allocation are jointly optimized [143]. Nonetheless, both platforms face intrinsic trade-offs: fixed-wing UAVs cannot hover and must maintain continuous motion, leading to fluctuating link quality, service fairness challenges, and increased deployment and optimization complexity, which often confines them to boundary or relay roles rather than hotspot servicing [110,140,142]. High-altitude UAVs, while benefiting from improved propagation conditions, incur increased free-space path loss and higher transmission power requirements as altitude rises, resulting in diminishing throughput returns beyond an optimal height and imposing energy and hardware cost constraints [142,143].

In synthesizing these findings, although Rotary-Wing Drones (RWDs), Fixed-Wing Drones (FWDs), and High-Altitude Platforms (HAPs) share the common objective of facilitating post-disaster communication, each type presents a unique trade-off profile that influences its feasibility in different scenarios. RWDs are constrained by high energy consumption, leading to short service times, and are best suited for rapid, localized communication restoration [154–156], though they require sophisticated deployment algorithms [6,31], or hardware modifications to be efficient at scale [157]. FWDs offer better energy efficiency and broader coverage, but are less effective in complex terrains such as dense urban or hilly areas [34,35]. Finally, HAPs provide exceptional endurance and advanced data processing capabilities, as validated by both theoretical models [35,158], but are confronted with high deployment costs, geographical dependencies, and strict airspace regulations, which limit their utility for rapid response. To summarize the capabilities of each type, a comparative analysis is provided in Table 4.

These platforms' characteristics highlight the limitations of homogeneous UAVs deployments and motivate heterogeneous, multi-tier ABS architectures for post-disaster communication. Rather than forcing a single platform to satisfy conflicting objectives, hybrid fleets allocate specialized roles according to flight mechanics and resource capabilities: RWDs provide hovering fronthaul access over user clusters, FWDs deliver high-endurance relay and perimeter backbone connectivity, and tethered UAVs, where available, offer quasi-permanent, high-capacity backhaul links supported by ground power [41,79,92]. This role-based separation effectively decouples coverage from connectivity, allowing optimization frameworks to expand service areas without compromising network stability [80,110]. Simulation and field-inspired studies consistently show that such heterogeneous architectures outperform single-type fleets in terms of coverage, resilience, and adaptability under dynamic disaster conditions [110]. Moreover, explicitly accounting for resource heterogeneity (such as differences in payload, endurance, and transmission power) enables more energy-efficient task allocation, for example by assigning FWDs to long-distance data transport and RWDs to localized hovering services [83,92,110]. Collectively, multi-UAV-type

Table 4
Summary of UAVs types and capabilities for post-disaster communication.

Criteria	UAVs type			Reference
	RWDs	FWDs	HAPs	
Coverage Scale	Localized coverage in confined or obstructed environments	Regional or corridor-wide coverage	Wide-area / district to regional-scale coverage	[50,141,143]
Endurance	Limited (Approx. 15 – 90 mins; battery dependent)	Extended (Hours to days via solar or high-efficiency cruise)	High endurance (Hours to days for solar-powered HAPs)	[63,92,143]
Terrain Sustainability	3D maneuverability; suitable for urban, mountainous, and rural debris	Unobstructed airspaces; suitable for large/inaccessible regions	Operates at adjustable altitudes to optimize LoS links and avoid terrestrial obstacles	[49,78,140]
Operational Strength	- Precision hover enables targeted, short-range connectivity.	- High efficiency supports persistent, mid-range coverage and relay.	- Persistent strategic backbone for long-duration communication.	[51,78,110,158]
Primary Limitation	Limited battery capacity, payload weight constraints, and susceptibility to wind.	Inability to hover, requiring linear flight at high altitudes, and sensitivity to turbulence.	- High deployment cost, regulatory and atmospheric constraints.	[35,63,78,110, 155,156,158]
Best-fit Scenario	Localized infrastructure collapse, earthquakes, landslides, and urban search and rescue.	Floods, wildfires, and large distributed disasters.	Long-term emergency coverage, remote/rural area connectivity, and wide-area data collection.	[51,92,143]

architectures could be a feasible design principle for scalable, energy-efficient, and resilient post-disaster aerial communication systems.

3.2.2. *Deployment and coordination strategies*

UAVs deployment strategies in post-disaster recovery operations can employ either a single UAVs or multiple UAVs, with the choice being contingent upon the scale of the disaster. A single UAVs may suffice for smaller-scale disaster scenarios, enabling the rapid restoration of communications within a localized area with minimal operational cost and complexity. However, UAVs fleets are generally preferred for large-scale disasters due to their inherent advantages in redundancy, broader coverage, and enhanced endurance. Centralized, decentralized, and hybrid are predominant deployment strategies to address diverse disaster situations [159]. These strategies are not mutually exclusive but represent a spectrum of trade-offs, balancing the need for optimal, global coordination against the demand for real-time, dynamic adaptability.

Centralized deployment strategies rely on a single coordinating entity, such as a ground station, cloud server, or super-domain controller, to aggregate global network information, including user equipment locations, channel state information, and UAVs energy levels, and to jointly optimize three-dimensional placement and trajectories for the entire fleet [98,104,132]. This global view enables coordinated decisions that are difficult to achieve through local reasoning alone. Recent advancements have enriched centralized approaches through the integration of Artificial Intelligence (AI), meta-heuristic optimization, and Software-Defined Networking (SDN). Centralized Deep Reinforcement Learning (DRL) methods (e.g., DDPG-based positioning and Q-learning frameworks) support continuous joint optimization of UAVs locations, power allocation, and user association, improving feasible operating regions and reducing outage compared with greedy heuristics [122, 146]. More advanced designs employ Multi-Agent Proximal Policy Optimization (MAPPO) with a centralized critic to enforce cooperative behavior while minimizing energy consumption [115]. In parallel, nature-inspired meta-heuristics—including JAYA, enhanced genetic algorithms, grey wolf optimization (with chaotic or K-means-based initialization), and UAV-artificial bee colony methods—have demonstrated improved convergence, placement accuracy, and latency performance when addressing the NP-hard nature of 3D deployment problems [26,45,49,65,74]. To better represent heterogeneous user distributions, centralized frameworks increasingly incorporate intelligent clustering techniques such as Gaussian mixture models, fuzzy C-means, and neural-guided hybrid clustering, which provide more realistic inputs for placement optimization and accelerate convergence in complex terrains [48,96,121]. SDN-based architectures further extend centralized control by separating control and data planes, introducing hierarchical controllers, and applying queueing-based packet processing to improve scalability and mitigate congestion and single-point-of-failure risks [147].

In contrast, decentralized deployment strategies treat each UAVs as an autonomous decision-making agent that relies on local observations and limited information exchange with neighboring UAVs, without dependence on a global controller. This paradigm emphasizes self-organization and local adaptability, and is commonly realized through Multi-Agent Reinforcement Learning (MARL). Q-learning, deep Q-networks, and continuous-control variants such as MASAC and DDPG enable UAVs to adjust their three-dimensional positions and trajectories to maximize coverage while preserving connectivity and minimizing service disruption [47,125]. Collaborative schemes such as CMAD-DDQN, proposed by Omoniwa, Galkin and Dusparic [120], further enhance coordination and energy efficiency by allowing UAVs to share local telemetry with nearby peers. Alongside learning-based methods, decentralized deployments also draw on virtual force and potential-field models, game-theoretic formulations, and distributed clustering algorithms to achieve collision avoidance, interference management, and load balancing using only local knowledge [91,97,114, 131,152]. These approaches distribute computational complexity across the fleet, support scalable operation, and enable rapid response to local environmental changes, while reducing control-plane signaling overhead [43,47,87,120]. At the same time, effective decentralized operation requires sufficient onboard computational resources, reliable short-range communication interfaces, local sensing of users and obstacles, energy-aware decision policies, and well-designed convergence mechanisms to ensure stable collective behavior under partial and dynamic information [87,88,105,131,151,152].

Bridging these two paradigms, hybrid deployment strategies combine centralized global coordination with decentralized local adaptation to exploit the strengths of both approaches. In typical designs, a centralized controller uses available global or quasi-static information, such as satellite imagery or historical call records, to compute an initial or coarse-grained fleet configuration via global optimization, after which individual UAVs perform decentralized local adjustments based on real-time sensing and evolving user demand [58]. This separation of roles limits the computational and signaling burden of continuous global optimization while avoiding poor local equilibria. Hybrid architectures also support hierarchical information processing under partial observability, where terrestrial edge agents respond to localized failures or traffic variations and UAVs agents periodically aggregate this knowledge to update global parameters, maintaining coherent network behavior even under intermittent backhaul connectivity [60,75]. From an AI perspective, hybrid deployment naturally aligns with Centralized Training and Decentralized Execution (CTDE), enabling complex DRL policies to be trained using global data and then executed autonomously onboard UAVs based on local observations [142]. Furthermore, hybrid frameworks facilitate role-based

optimization, in which centralized planning enforces global connectivity and backhaul robustness, while decentralized controllers guide coverage-oriented UAVs toward local service objectives, jointly supporting wide-area coverage and resilient network connectivity in disaster-stricken environments [80]. The following Table 5 summarizes centralized, decentralized, and hybrid UAVs deployment strategies for communication restoration.

3.2.3. AI-driven UAVs operations

Traditional UAVs operations often rely on pre-programmed flight paths or manual control, which lack the adaptability required for highly dynamic and unpredictable environments. This limitation has created a significant opportunity for advanced technologies like AI to enable autonomous, real-time decision-making. This techniques offer novel solutions in addressing the inherent challenges posed by dynamic environments, uncertain user distributions, and severe energy constraints. In particular, AI-based trajectory and path optimization reframes UAVs mobility control as a learning problem rather than a static planning task. DRL models the trajectory design as a Markov Decision Process (MDP), enabling UAVs to learn motion policies through continuous interaction with the environment [52,63]. Value-based methods such as Deep Q-Networks (DQN) jointly optimize flight paths and communication performance by balancing throughput rewards against operational penalties [52], while constrained formulations such as Safe-DQN incorporate Lyapunov-based mechanisms to guarantee safety and energy compliance [63]. Policy-gradient approaches, including Proximal Policy Optimization (PPO), further improve learning stability in continuous action spaces and, when combined with Channel Knowledge Maps (CKM), produce trajectories that are robust to positioning errors and channel uncertainty in complex disaster terrains [94]. In multi-UAV scenarios, transformer-enhanced MAPPO frameworks enable parallel task execution and coordinated global optimization without restrictive action-space assumptions, significantly improving scalability and convergence [130]. Complementary to learning-based methods, meta-heuristic algorithms—such as Genetic Algorithms (GA), Particle Swarm Optimization (PSO), and Grey Wolf Optimization (GWO)—remain effective for exploring highly non-convex solution spaces, where clustering-assisted initialization and evolutionary rule learning help avoid local optima and adapt trajectories to heterogeneous rescue priorities [26,131,160]. For environments with limited or uncertain prior information, Multi-Armed Bandit (MAB) formulations further balance exploration and exploitation across spatial grids while explicitly accounting for UAVs and user energy costs [105]. Recent hybrid frameworks that integrate global evolutionary search with local Reinforcement Learning (RL) or neural-guided clustering have demonstrated improved robustness, coverage, and convergence speed in realistic disaster scenarios [110,121].

Beyond trajectory optimization, AI has driven substantial progress toward autonomous control and network-level intelligence in UAV-based ABSs deployments. Contemporary systems increasingly adopt CTDE, as exemplified by MAPPO and MADDPG, allowing UAVs to learn globally coordinated policies while making independent real-time decisions based on local observations—an essential property when centralized supervision or backhaul connectivity is unavailable after disasters [60,88,115]. Distributed collaboration mechanisms, such as CMAD-DDQN, further enhance autonomy by enabling neighboring UAVs to exchange telemetry and coordinate coverage and mobility to improve energy efficiency and network connectivity [47]. Hierarchical learning frameworks extend this paradigm by integrating air–ground cooperation, where terrestrial edge nodes infer local environmental statistics and UAVs agents aggregate this knowledge to autonomously optimize hovering and relay positions [120]. AI-based control also enables joint, real-time optimization of trajectories, user association, and power allocation using DRL agents (e.g., SAC, DDPG), allowing UAVs to simultaneously improve LoS, reduce interference, and conserve energy [62,108]. Priority-aware handover and user association strategies further ensure that emergency traffic is served preferentially over purely signal-strength-driven policies [106], while integrated access and backhaul (IAB) optimization maintains connectivity to the core network even when macro base stations are damaged [87].

Crucially, AI-driven ABS control is inherently designed to adapt to uncertainty, failures, and environmental dynamics, treating UAVs operation as a continuous decision-making process rather than a fixed optimization problem. Through persistent state observation and reward-driven policy updates, DRL agents can rapidly respond to sudden demand surges, user mobility, or channel degradation by adjusting positions and transmission parameters in real time [60,142]. Decentralized MARL frameworks such as MASAC and CMAD-DDQN enable self-organization under partial information, allowing neighboring UAVs to compensate for node failures, battery depletion, or attacks and to restore coverage without centralized coordination [47,80,120]. Advanced neural architectures, including attention-enhanced LSTM models and Graph Neural Networks (GNNs), further improve temporal and spatial awareness, enabling UAVs to focus on unserved user clusters and to operate reliably under rapidly changing network topologies [88,113]. To accelerate adaptation in unseen disaster scenarios, curriculum learning and federated reinforcement distillation facilitate faster convergence by transferring prior knowledge while reducing

Table 5
UAVs deployment strategies for post-disaster communication recovery.

Strategy	Description	Context	Advantages	Limitations	Reference
Centralized	The central controller uses global data (user locations, environment) to plan UAVs placement.	- Serving areas with high user density (e.g., parks, grounds, rooftops).	- Maximizes the number of serviced users and user connectivity. - Minimizes the number of UAVs required and network deployment costs. - Improves connection throughput and signal quality.	- High computational complexity, increasing exponentially with network size. - Difficulty in obtaining timely and accurate global environmental and user mobility information in dynamic disaster scenarios.	[23,46,50,51,56]
Decentralized	UAVs autonomously adjust positions using local sensor or neighbor data, with no central control.	- Static or simulated environment where network state information is known in advance.	- Easily scalable as the number of UAVs or users increases. - The system is more resilient to failures of a single UAVs. - Adapts and responds promptly to dynamic changes in the environment and user location.	- Requires large signal exchange, causing bandwidth overload and transmission delays. - Difficult to scale, less stable as the number of UAVs or users increases. - Inflexible in response to dynamic changes in the environment and user location.	[47,88,105,125,151]
Hybrid	Combines centralized initial planning with decentralized UAVs adjustments.	- Complex disaster-hit areas require both planning and adaptability.	- Balances global optimization with local adaptability. - Suits complex, dynamic scenarios.	- Inherits centralized planning time and decentralized communication needs. - Difficult to adapt in real time to dynamic environmental changes.	[58,60,80,141,142]

communication overhead [47,150]. Although most of these strategies have been validated through simulations—demonstrating substantial gains in coverage, fairness, energy efficiency, and adaptability over traditional heuristics—emerging experimental evaluations and field-based studies provide encouraging evidence of their practical feasibility. Collectively, these advances position AI-based optimization as a robust and scalable foundation for fully autonomous, resilient, and energy-aware UAV-based ABS deployments in post-disaster environments.

3.2.4. Energy optimization for endurable UAV-based ABSs

UAV-based ABSs deployed for post-disaster communication operate under onboard energy constraints, which has driven various research into energy optimization strategies spanning mobility control, network coordination, and physical-layer design. Energy conservation strategies primarily aim to reduce consumption without introducing additional power sources. At the platform level, intelligent trajectory and speed optimization replace static placement with adaptive mobility control, where speed-aware cruising, spiral-path planning, and energy-penalized DRL jointly minimize propulsion energy while maintaining coverage objectives [40,45,115,132]. DRL further improves efficiency by preventing redundant flight paths among multiple UAVs, yielding substantial savings in energy consumption [115]. These methods, however, rely on accurate real-time environmental awareness and incur non-negligible computational overhead, which may limit their robustness under chaotic post-disaster conditions [45,47,128]. At the network level, collaborative resource management and clustering strategies shift the focus from individual UAVs optimization to fleet-wide efficiency. Sleep-mode protocols deactivate redundant UAVs to extend overall network lifetime, Device-to-Device (D2D) offloading reduces reliance on energy-intensive UAVs links and can cut total power consumption by more than 50% according to He, Wang [145], and energy-aware clustering avoids assigning highly mobile nodes as cluster heads to stabilize energy usage [97]. While effective, these schemes introduce signaling overhead and require careful interference management, particularly in dense D2D scenarios [76,145]. Complementary techniques such as semantic communication further reduce energy demand by transmitting extracted meaning instead of raw data, thereby shortening transmission time and indirectly lowering hovering-related propulsion energy, albeit at the cost of additional onboard processing [70]. At the physical layer, non-LoS free-space optical communication (NLoS-FSOC) provides an alternative to radio-frequency links, achieving orders-of-magnitude higher energy efficiency through optical broadcast via diffuse reflectors, but requiring specialized hardware and favorable deployment conditions [64].

In parallel, energy harvesting strategies seek to supplement or replace onboard batteries, trading flexibility for endurance. Namely, tethered UAVs represent a reliable harvesting-based solution, supplying continuous power and high-capacity wired backhaul from ground generators to enable long-duration operation and support heavy communication payloads, albeit with significantly constrained mobility and deployment flexibility [41,86,135,139]. Wireless harvesting alternatives preserve aerial mobility but offer more limited and environment-dependent performance. Optical Wireless Power Transfer (OWPT) and laser charging achieve higher conversion efficiency than RF harvesting and can extend hovering time under clear LoS conditions, yet their performance degrades in adverse weather and remains largely validated through simulation studies [86,119]. RF energy harvesting provides only low-level supplemental power, typically for relay or cluster-head UAVs, but can modestly prolong network lifetime with minimal hardware overhead [76,119]. Solar energy harvesting, most effective for FWDs with large surface areas, can significantly extend endurance during daylight hours, but remains insufficient for power-intensive rotary-wing platforms and is strongly constrained by weather and available panel area [86,135].

Given the intrinsic limitations of both conservation-only and harvesting-only solutions, hybrid energy strategies have emerged as a more resilient paradigm for post-disaster ABS deployments. These approaches tightly integrate energy acquisition with mobility planning, network topology control, and communication-layer optimization. For example, solar-assisted FWDs combined with energy-aware trajectory planning and gravitational energy storage can approach near-continuous operation under favorable environmental conditions [89]. At the fleet level, schemes such as ICBM-UAV exploit dual power sources alongside clustering and adaptive sleep mechanisms to suppress redundant activity and extend overall network lifetime [116]. At the communication layer, collaborative D2D/D2I resource management further reduces transmission energy through joint spectrum and power optimization, albeit with increased algorithmic complexity [145]. Together, sustainable UAV-based ABSs might require more than isolated techniques; instead, they demand integrated, cross-layer energy optimization frameworks that jointly balance endurance, coverage, and operational robustness in dynamic post-disaster environments.

4. Discussion

As outlined in the methodology, this paper utilized a systematic literature review following the PRISMA protocol to synthesize research on applying UAVs as ABSs in post-disaster scenarios. Eligible studies were identified and analyzed across the key dimensions of this research domain. However, an in-depth examination reveals that despite growing interest, the field continues to face substantial limitations and challenges that constrain its effective real-world implementation.

4.1. Challenges and barriers

When deployed as temporary substitutes for non-operational terrestrial base stations, UAV-based ABSs should meet service continuity, coverage reliability, and backhaul robustness requirements comparable to those of ground infrastructure, yet under far more constrained operating conditions. This contrast exposes several fundamental challenges that continue to limit their transition from experimental solutions to dependable emergency communication assets. Most critically, energy constraints and limited endurance remain the dominant bottleneck, as UAVs must simultaneously sustain flight and radio access functions that are normally powered by stable terrestrial grids. Unlike fixed base stations, UAV-ABSs must continuously trade off propulsion energy against communication energy, where prolonged hovering and high-throughput transmission impose substantial power demands and frequently interrupt service due to battery depletion [40,77,119,130]. This limitation is further intensified by payload–battery coupling, since increasing onboard energy storage directly reduces the payload available for baseband processing and radio hardware, and by disaster-induced power outages that impede both UAVs recharging and end-user device survivability. In parallel, backhaul connectivity, a function inherently stable in terrestrial networks, becomes a fragile dependency for UAV-ABSs. Even when access links to ground users are strong, reliance on satellite or long-distance wireless backhaul often introduces severe capacity constraints, undermining end-to-end performance and limiting the ability of UAVs to emulate terrestrial base station functionality [82,119,139]. Maintaining high-quality LoS backhaul, particularly for mmWave or Free Space Optical (FSO) links, is further complicated by UAVs motion, atmospheric conditions, and urban obstructions, resulting in intermittent or unstable connectivity [86,118]. Moreover, dynamic three-dimensional deployment and trajectory planning impose algorithmic challenges absent from fixed terrestrial networks, as UAVs must continuously reposition to balance coverage, interference, and energy consumption in environments where user locations are unknown, mobility is high, and higher altitudes simultaneously improve LoS and exacerbate co-channel interference [26,40,48,49,65,102,122,125]. Finally, scalability and coordination requirements emerge when multiple UAVs jointly replace damaged base stations, demanding tight

synchronization for collision avoidance, interference control, and load balancing. While centralized control can approximate terrestrial network planning, it is highly vulnerable to backhaul disruptions, whereas decentralized coordination improves resilience but requires sophisticated mechanisms to avoid coverage fragmentation and service overlap [43,80,87,147].

Beyond these architectural and operational challenges, a critical gap persists between theoretical feasibility and operational readiness. Most existing studies validate UAV-based ABSs through simulation under idealized assumptions, such as non-depleting batteries, perfect channel state information, or uninterrupted wireless charging, which only partially reflect the realities of disaster environments. Although such evaluations provide important conceptual insights, they fall short of demonstrating whether UAV-ABSs can sustain carrier-grade reliability comparable to terrestrial base stations under unpredictable weather, hardware imperfections, mobility uncertainty, and fluctuating demand. Moreover, despite the demonstrated benefits of RL-based control approaches, their practical deployment in post-disaster scenarios remains constrained by computational latency and hardware limitations. Most DRL and MARL frameworks were trained and evaluated using high-performance GPU-enabled testbeds, whereas UAVs flight controllers employ limited embedded processors that cannot support complex real-time inference workloads [16,17]. This highlights a crucial gap between theoretically robust models, and hardware-in-the-loop reality, as mentioned in the study of Zhou, Liu [161]. Specifically, the work by Zhang, Lu [162], while not an RL model, directly confronts this by implementing an LTE base station on an NVIDIA Jetson TX2 airborne computer. Their findings which show higher latency and lower throughput compared to a computer-based system perfectly illustrate the real-world processing bottlenecks that complex, non-optimized DRL policies would face. As a result, real-time policy execution can introduce decision delays exceeding safe operation thresholds in rapidly changing environments, particularly during dynamic obstacle avoidance or target re-localization [158]. This challenge is further amplified in multi-agent networks, where maintaining policy coherence requires continuous inter-UAV communication, increasing latency and energy overhead [47]. Hence, the scarcity of field trials and documented real-world disaster deployments therefore remains a major limitation, leaving the practical effectiveness of UAV-based ABSs as true temporary replacements for terrestrial infrastructure largely unproven. Bridging this gap will require systematic experimental validation and mission-scale trials that assess not only coverage and throughput, but also endurance, backhaul stability, coordination robustness, and service continuity under realistic post-disaster conditions.

Additionally, widespread UAVs adoption is also hindered by significant non-technical. A primary challenge is the lack of clear, standardized regulatory frameworks, these regulatory barriers can delay rapid response operations. Furthermore, the use of UAV-ABSs raises significant privacy concerns and security issues. This creates complex ethical dilemmas, such as the need to balance response urgency with individual privacy rights. These concerns directly impact community acceptance, which is essential for successful deployment in populated areas [16,17,163,164].

4.2. Future research agenda

Advancing UAV-based ABSs from simulation-driven concepts to reliable temporary replacements for non-operational terrestrial infrastructure requires a shift toward rigorous validation, deployable intelligence, and system-level co-design. A primary research direction is the development of high-fidelity validation frameworks that combine detailed physical models of UAVs battery degradation, propulsion energy consumption, sensor characteristics, and flight dynamics with advanced radio propagation simulations capturing path loss, shadowing, blockage, mobility-induced fading, and inter-UAV interference. These simulation environments should be complemented by real-world and semi-synthetic datasets, including field measurements and hardware-in-the-loop experiments, to bridge the gap between idealized assumptions and operational realities. Such integrated validation approaches would enable reproducible, end-to-end assessment of heterogeneous UAVs platforms, AI control algorithms, and energy management strategies under disaster-relevant conditions, thereby mitigating the current over-reliance on purely theoretical or laboratory-based evaluations.

Building upon robust validation, a second priority is the design of lightweight and secure AI algorithms suitable for execution on resource-constrained UAVs hardware. Future work should focus on model compression techniques (e.g., quantization, pruning, and knowledge distillation) to reduce computational and energy overhead while preserving control performance. In parallel, security and privacy must be embedded into AI design through adversarial training and robustness-aware learning to mitigate threats such as sensor manipulation and GPS spoofing. Federated learning frameworks further offer a promising pathway to collaborative model training without centralized data sharing, which is particularly critical in sensitive disaster-response scenarios.

A third research direction concerns the integration of hybrid energy systems and multi-layer network architectures to sustain long-duration operation at scale. Promising designs combine solar harvesting, high-performance batteries, and wireless power transfer, alongside layered deployments in which RWDs serve local hotspots, FWDs provide wide-area backbone connectivity, and HAPs or satellites ensure persistent backhaul. Embedding predictive, energy-aware AI into these architectures will be essential for proactive resource allocation under uncertain demand and environmental conditions.

Finally, realizing large-scale and cross-border deployment requires the establishment of standardized and ethically grounded frameworks. Harmonized technical standards, interoperable control interfaces, and agile regulatory sandboxes are needed to accelerate adoption and ensure operational safety. At the same time, ethical considerations must be explicitly incorporated into system and AI design, given the life-critical role of UAV-based ABSs in disaster recovery. Collectively, these directions outline a potential pathway for transitioning UAV-based ABSs from theoretical feasibility to operational maturity and broader societal trust (Fig. 3).

5. Conclusions

This study provides a comprehensive review and thematic synthesis of 117 peer-reviewed publications on using UAV-based ABSs in post-disaster scenarios. Leveraging the PRISMA methodology, the review conducts an in-depth analysis of research trajectories, regarding four core aspects: UAVs Types and Capabilities, Deployment and Coordination Strategies, AI-Driven UAVs Operations, and Energy Optimization. Through this synthesis, the study illuminated the current knowledge state while highlighting significant concerns and identifying potential avenues for future research.

To date, UAV-based frameworks for ABSs deployment have evolved from isolated technological solutions into increasingly cohesive, adaptive, and intelligent systems, demonstrating considerable feasibility for emergency communication support. Different UAVs platforms, namely RWDs, FWDs, and HAPs, exhibit distinct performance trade-offs, indicating that coordinated and heterogeneous deployment strategies are most suitable for large-scale disaster management. RL has emerged as the principal AI paradigm for enhancing UAVs operations in complex and dynamic environments, with advanced variants such as DRL and MARL enabling greater autonomy, adaptability, and efficiency. However, several critical challenges remain, particularly those related to limited endurance, stringent energy constraints, and regulatory compliance, which continue to hinder the sustainable deployment of UAV-based ABSs in post-disaster scenarios. In addition, the scarcity of field trials and real-world case studies represents a major barrier to

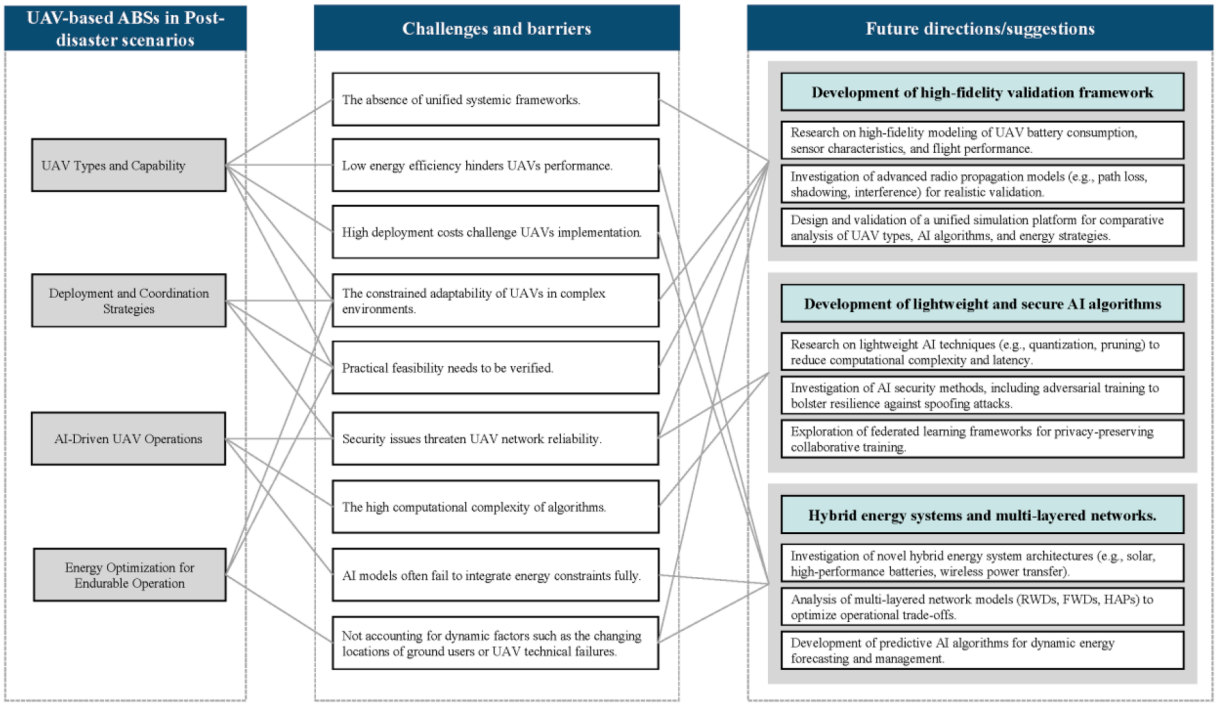


Fig. 3. Challenges, barriers, and future directions related to UAV-based ABSs.

advancing these systems from theoretical feasibility toward operational readiness. Taken together, these observations highlight both the significant progress achieved and the important theoretical and practical contributions of this study, while also delineating clear directions for future research and deployment.

Nevertheless, the study's limitations should be acknowledged. Firstly, the use of a manual screening approach in this systematic review, rather than specialized bibliometric software, may have resulted in the omission of some relevant studies. However, the authors conducted careful and iterative checks throughout the screening process to minimize this possibility. Secondly, excluding industrial reference sources such as market reports, trade magazines, and national standards constrains the comparison between academic research and real-world applications. To mitigate these limitations, future research should broaden its scope to include additional databases and industry literature while incorporating qualitative studies to bridge better the gap between theoretical advancements and practical implementation challenges.

Relevance to resilience

Rapid restoration of communication recovery is a critical component, essential for effective coordination, emergency response, and resource allocation in post-disaster scenarios. The findings of this review directly contribute to strengthening this adaptive capacity by providing a comprehensive analysis of current operational capabilities, deployment models, and systemic bottlenecks. By ensuring the continuity of communication for first responders and affected populations, these systems are a critical component in enhancing the overall resilience of disaster management efforts. Ultimately, this synthesis helps bridge the gap between theoretical advancements and practical implementation, guiding the development of more robust, efficient, and reliable UAV-based solutions for future crises.

Declaration of generative AI and AI-assisted technologies in the writing process

During the preparation of this work, the authors used ChatGPT for proofreading and for generating selected elements of the UAV-based ABS concept illustration in the study's graphical abstract. After using this tool, the authors reviewed and edited the content as needed and take full responsibility for the content of the published article.

Data availability

Data will be made available on request.

CRediT authorship contribution statement

Raymond Thapa Magar: Writing – original draft, Visualization, Software, Methodology, Investigation, Data curation, Conceptualization. **Tien Phat Dinh:** Writing – original draft, Visualization, Software, Methodology, Investigation, Data curation. **The Hung Nguyen:** Data curation, Validation, Visualization. **Quang Hoai Le:** Writing – review & editing, Validation, Supervision, Project administration, Methodology. **Ahn Yonghan:** Validation, Supervision, Project administration, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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